

Orchid Majumder

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WORK EXPERIENCE

Amazon AGI

Seattle, USA

Principal Research Engineer, Amazon Nova (Multi-Modal Pre-Training)

October 2023 – current

- Research engineering lead for Amazon Nova foundational models pre-training. Developed a *NeMo-Megatron* based distributed-training stack that is used to train Nova models (*using 5D parallelism*).
- Working on architecture research and pre-training scaling laws for Nova-2 and beyond.

AWS AI Labs

Seattle, USA

Senior Research Engineer, Diffusion Models / Computer Vision

April 2019 – September 2023

- Research engineering lead for Amazon Titan multi-modal models and text-to-image/video diffusion models. Developed a *PyTorch-FSDP* based distributed-training stack that was used to train large-scale image/video diffusion models on AWS.
- Research engineering lead for Amazon Rekognition Custom Labels, an AWS AutoML service aimed towards helping customers build their own ML models with a few labeled samples.
- Worked on research topics related to few-shot, continual and meta-learning algorithms for visual recognition and hyperparameter optimization (*see [Google Scholar](#)*).

AWS SageMaker

Seattle, USA

Research Engineer, AI Labs

March 2017 – March 2019

- Part of the core team that launched Amazon SageMaker at AWS re:Invent 2017, a fully managed AWS service for building, training and deploying any kind of ML models.
- Led the development effort for [Inference Pipeline](#) for Amazon SageMaker.
- Developed deep-learning based NLP algorithms including [Seq2Seq](#) and [Object2Vec](#) which are offered as built-in algorithms within Amazon SageMaker.

Amazon

Bangalore, India

Software Engineer, Amazon India

July 2014 – March 2017

- Engineering lead for registration and order management for individual sellers on [Amazon India](#) — led a large-scale cross-team effort to develop a software system that enabled individual sellers to get started as professional sellers.
- Built a rule-based recommendation engine to help sellers determine the price for their products, repurposed Amazon logistics stack for individual sellers and simplified the registration workflow for new individual sellers.

INTERNSHIP EXPERIENCE

Amazon

Bangalore, India

Software Engineer Intern, Amazon Retail

June 2013 – July 2013

- Built an intermediate caching layer using Amazon DynamoDB to reduce request latency.
- Built a distributed log parser using Hadoop/MapReduce on Amazon EMR.

Indian Statistical Institute

Kolkata, India

Research Intern, Machine Perception / Computer Vision

June 2012 – July 2012

- Worked on fuzzy neural-network based algorithms to identify Indian dance styles from images.

SELECTED PUBLICATIONS

The Amazon Nova Family of Models: Technical Report and Model Card (arXiv) – Amazon AGI.

On the Scalability of Diffusion-Based Text-to-Image Generation (CVPR 2024) – Hao Li, Yang Zou, Ying Wang, Orchid Majumder, Yusheng Xie, R Manmatha, Ashwin Swaminathan, Zhuowen Tu, Stefano Ermon, Stefano Soatto.

Revisiting Contrastive Learning for Few-Shot Classification (arXiv) – Orchid Majumder, Avinash Ravichandran, Subhransu Maji, Marzia Polito, Rahul Bhotika, Stefano Soatto.

Incremental Meta-Learning via Indirect Discriminant Analysis (ECCV 2020) – Qing Liu, Orchid Majumder, Avinash Ravichandran, Rahul Bhotika, Stefano Soatto.

MARTHE: Scheduling the Learning Rate Via Online Hypergradients (IJCAI 2020) – Michele Donini, Luca Franceschi, Orchid Majumder, Massimiliano Pontil, Paolo Frasconi.

d-SNE: Domain Adaptation Using Stochastic Neighborhood Embedding (CVPR 2019 Oral) – Xiang Xu, Xiong Zhou, Ragav Venkatesan, Gurumurthy Swaminathan, Orchid Majumder.

CONFERENCES

Area Chair – NeurIPS, CVPR, ECCV.

Outstanding Reviewer – NeurIPS, CVPR, ECCV.

Peer Review – NeurIPS, ICLR, CVPR, ICML, ECCV, ICCV.

Journal Review – TPAMI, TMLR, IEEE TCSVT.

PATENTS

Automated Machine Learning Pipeline Exploration and Deployment (US : 11727314B2).

Building Sequence to Sequence (S2S) models using Previously Generated S2S Models (US : 11748610B1).

ACADEMICS

Columbia University in the City of New York

M.S. in Computer Science (Machine Learning Track)

New York, USA

2020 – 2022

Jadavpur University

B.S. in Computer Science and Engineering

Kolkata, India

2010 – 2014

OPEN SOURCE

adatune | [GitHub](#)

- Implementation of IJCAI 2020 paper - “MARTHE: Scheduling the Learning Rate Via Online Hypergradients”.

sagemaker-sparkml-serving-container | [GitHub](#)

- A toolkit for deploying SparkML models efficiently in SageMaker without an Apache Spark runtime.

OTHER ACHIEVEMENTS

Mathematics Olympiad – One of the 500 students all over India to qualify for the Indian National Mathematical Olympiad (INMO) in 2008.

Entrance Examinations – Ranked in the top 0.1% in the secondary examination, higher-secondary examination and engineering entrance examination in the state of West Bengal in India.